

Group counting expressions as a window into the internal structure of numerals

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July 8, 2026



This talk

Abstract use of numerals

(1) **Two** is my favorite number.

Counting use of numerals

(2) Eve ate **two** apples.

Previous work: numeral counting expressions like (2) properly contain abstract number concept expressions like (1). (Wągiel and Caha 2020; Wągiel and Caha 2021)

This talk: Also number concept expressions like (1) are **internally complex**.

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This talk: Also number concept expressions like (1) are **internally complex**.

Data

- ▶ Understudied numeral constructions: **group counting** expressions

(3) **three sets** of cards

- ▶ Taking into account group counting constructions yields paradigms that provide insights into internal make-up of numerals.

⇒ Our findings inform debates on the relation between abstract number concepts, unit and group counting in natural language. (E.g. Bale, Gagnon, et al. 2011; Bale and Coon 2014; Bale, Coon, and Arcos 2019; Ionin and Matushansky 2006; Grimm and Dočekal 2021; Sağ 2018; Rothstein 2017)

Roadmap

Unit vs. group counting

Pattern 1: unit $\rightarrow$ bimorphemic

Pattern 2: unit $\rightarrow$ bimorphemic

group $\rightarrow$ bimorphemic

group $\rightarrow$ tri-morphemic

Abstract use of numerals

Abstract number concepts

Pattern 1: abstract bare *Num*

Pattern 2: abstract *Num+A*

Summary of patterns

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Mandarin and Uzbek

Yucatec Maya and Bwamu

Conclusion

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Unit vs. group counting

Unit counting use of numerals

- ▶ By far most studied type of numeral construction
- ▶ Numerals counting **NP-atoms**
- ▶ (4): set of pluralities each consisting of five apple atoms

(Landman 2003; Krifka 1989)

(4) **five** apples

Unit vs. group counting

Unit counting use of numerals

- ▶ By far most studied type of numeral construction
- ▶ Numerals counting **NP-atoms**
- ▶ (4): set of pluralities each consisting of five apple atoms
(Landman 2003; Krifka 1989)

(4) **five** apples

Group counting use of numerals

- ▶ Counting of **NP-non-atoms**
- ▶ (5): set of pluralities each consisting of five woman non-atoms
(Landman 1989; Dočekal 2012; Grimm and Dočekal 2021; Khrizman 2020)

(5) **five groups of** women

- ▶ Other languages: dedicated group counting numerals or classifiers
 - $\text{Num}_{\text{gr}} + \text{NP}$ ▪ $\text{Num} + \text{CLF}_{\text{gr}} + \text{NP}$

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Unit vs. group counting: Pattern 1

Mandarin

- (6) a. sān **tóu** niú
three CLF₁ cow
'three cows'
- b. sān **qún** niú
three CLF₂ cow
'three herds of cows'

Uzbek

- (7) a. ikki **dona** kitob
two CLF₁ book
'two books'
- b. ikki **guruh** bola
two CLF₂ child
'two groups of children'

Unit vs. group counting: Pattern 1

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 three CLF₁ cow
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Unit vs. group counting: Pattern 1

Pattern 1

Unit counting: **Num** + **A** 2-morphemic

Group counting: **Num** + **B** 2-morphemic

- ▶ Seems to be prevalent pattern in most classifier languages
- ▶ E.g. Chinese *three*, Uzbek *two*

NB: Not all numerals necessarily behave the same within a language.

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Unit vs. group counting: Pattern 2

Bwamu (Gur)

- (8) a. livru-ʒε **bo** tĩ
book-PL CLF₁ three
'three books'
- b. hã-ʒε **gnũ-nε** **bo** tĩ
woman-PL CLF₂-PL CLF₁ three
'three groups of women'

Yucatec Maya

- (9) a. jum-p'éeł kib
one-CLF₁ candle
'one candle'
- b. jun-p'èel **cháach** iib
one-CLF₁ CLF₂ bean
'one bunch of beans'

Unit vs. group counting: Pattern 2

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Unit vs. group counting: Pattern 2

Finnish

- (10) a. **vii-si** saapasta
five-SFX boot.PTV
'five-SFX₁ boots'
- b. **vii-de-t** saappaat
five-SFX₁-SFX₂ boots
'five pairs of boots'

Czech

- (11) a. **tř-i** karty
three-SFX cards
'three cards'
- b. **tr-oj-e** karty
three-SFX₁-SFX₂ cards
'three sets of cards'

Unit vs. group counting: Pattern 2

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- (10) a. NUM A
 vii-**si** saapasta
 five-SFX boot.PTV
 'five-SFX₁ boots'
- b. vii-**de-t** saappaat
 five-SFX₁-SFX₂ boots
 'five pairs of boots'

Czech

- (11) a. tř-i karty
 three-SFX cards
 'three cards'
- b. tr-**oj-e** karty
 three-SFX₁-SFX₂ cards
 'three sets of cards'

[Decomp. of *-oj - e* and *-de-t*: motivation in Appendix]

Unit vs. group counting: Pattern 2

Finnish

- (10) a. NUM A
 vii-**si** saapasta
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- b. NUM B C
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[Decomp. of *-oj* - *e* and *-de-t*: motivation in Appendix]

Unit vs. group counting: Pattern 2

Pattern 2

Unit counting:

Num + **A**

2-morphemic

Group counting:

Num + **A** + **C**

3-morphemic

(Bwamu, Y.Maya)

Num + **B** + **C**

3-morphemic

(Czech, Finnish)

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Abstract use of numerals

- ▶ Focus in semantic literature: numerals in their **counting function**
(e.g. Bale, Gagnon, et al. 2011, Martí 2020a; Martí 2020b, Scontras 2013, Saĝ 2018)
- ▶ Numerals with NP complements: **Num + NP**.
- ▶ Device for counting individuals in the NP denotation.
- ▶ Pluralities of NP individuals.

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Abstract use

- ▶ Non-counting contexts, e.g. maths, reference to number concepts
- ▶ English: only **bare numerals**

- (12) a. Four times two is eight.
 b. My favorite number is one.

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- (12) a. **Four** times **two** is eight.
b. My favorite number is **one**.

- ▶ Even semantically inert nouns cannot occur here
- ▶ Shows: these contexts select for items denoting **abstract number concepts**
(e.g. Rothstein 2017; see also Moltmann 2013; Moltmann 2017; Wągiel and Caha 2020; Wągiel and Caha 2021)

- (13) a. #**Four** things times **two** things is **eight** things.
b. #My favorite number is **one** thing.

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- (13) a. #**Four** things times **two** things is **eight** things.
b. #My favorite number is **one** thing.

- ⇒ Semantics of numerals is **not inherently** that of **counting** NP denotations
- ⇒ Certain instances of numerals denote **abstract number concepts**, not counting devices.

We will show

Pattern 1

Mandarin, Uzbek

Unit: **Num** + **A** 2-morph

Group: **Num** + **B** 2-morph

Abstract use

Num

Pattern 2

Bwamu, Y.Maya, Czech, Finnish

Unit: **Num** + **A** 2-morph

Group: **Num** + **A** + **C** 3-morph

Num + **B** + **C**

Abstract use

Num + **A**

We will show

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Unit: **Num** + **A** 2-morph

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Num

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Bwamu, Y.Maya, Czech, Finnish

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Group: **Num** + **A** + **C** 3-morph

Num + **B** + **C**

Abstract use

Num + **A**

⇒ What appears as **Num** in Pattern 1 realizes **two** features.

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Pattern 1: abstract bare *Num*

Mandarin

- (14) a. **jiǔ** chùyǐ **sān** shì **sān**.
nine divide.by three COP three
'Nine divided by three is three.'
- b. #**jiǔ** **gè** chùyǐ **sān** **gè** shì **sān** **gè**.
nine **CLF** divide.by three **CLF** COP three **CLF**

Uzbek (Levy-Forsythe 2025, p. 203)

- (15) a. **Ikki** karra **uch** barobar **olit**.
two times three equal six
'Two times three equals six.'
- b. #**Ikki-ta** karra **uch-ta** barobar **olit-ta**.
two-**CL_{unit}** times three-**CL_{unit}** equal six-**CL_{unit}**

- ⇒ No classifiers in abstract use of numerals
- ⇒ Classifiers in these languages are for **counting**

Full pattern 1

Pattern 1

Abstract:	Num	1-morphemic
Unit counting:	Num + A	2-morphemic
Group counting:	Num + B	2-morphemic

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Pattern 2: abstract *Num* + *A*

Bwamu (Bouwèrè Bognana, p.c.)

- (16) a. **bo** na la **bo** tĩ le **bo** hognu
CLF four PRE CLF three COP CLF seven
'Four plus three equals seven.'
- b. #na la tĩ le hognu
four PRE three COP seven

Yucatec Maya (Blaha Pfeiler and Skopeteas 2024)

- (17) a. **jum-p'éel** y-éetel u láak' ka'a-p'éel-e' óox-p'éel.
one-CLF_{unit} Ø-with A.3 other two-CLF_{unit}-TOP three-CLF_{unit}
'One and two is three.'
- b. #jum y-éetel u láak' ka'a-e' óox.
one-CLF_{unit} Ø-with A.3 other two-CLF_{unit} TOP three-CLF_{unit}

- ⇒ Numerals denoting abstract number concepts are **bi-morphemic**.
⇒ They need to contain a 'classifier'.

Pattern 2: abstract *Num* + *A*

Czech

- (18) a. Tř-i je prvočíslo.
three-SFX is prime.number
'Three is a prime number.'
- (19) a. #Tř-∅ je prvočíslo.
three-∅ is prime.number

Finnish (Karlsson 2017 p.300)

- (20) kak-si plus kak-si on neljä
two plus two is four
'two plus two is four'
- (21) a. #kak-∅ plus kak∅ on neljä-t
two-∅ plus two-∅ is four-PL

⇒ Numerals denoting abstract number concepts are **bi-morphemic**.

Full Pattern 1 and 2

Pattern 1

Abstract:	Num	1-morphemic
Unit counting:	Num + A	2-morphemic
Group counting:	Num + B	2-morphemic

Pattern 2

Abstract:	Num + A	2-morphemic	
Unit counting:	Num + A	2-morphemic	
Group counting:	Num + A + C	3-morphemic	(Bwamu, Y.Maya)
	Num + B + C	3-morphemic	(Czech, Finnish)

⇒ **Abstract number concept** denoting numerals are **internally complex** and composed of two features.

⇒ This complexity is **covert in Pattern 1** but **overt in Pattern 2**.

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Pattern 1 summary

Mandarin 'three'

san

san **tóu**_{unit}

san **qún**_{group}

Num

Num + **A**

Num + **B**

abstract

unit

group

Uzbek 'two'

ikki

ikki **dona**_{unit}

ikki **guruh**_{group}

Num

Num + **A**

Num + **B**

abstract

unit

group

Pattern 2 summary

Bwamu 'three'

bo tĩ

bo tĩ

gnũ-ne bo tĩ

Num + A

Num + A

Num + A + B

abstract

unit

group

Yucatec Maya 'one'

jum-p'éel

jum-p'éel

jum-p'éel cháach

Num + A

Num + A

Num + A + C

abstract

unit

group

Czech 'three'

tř-i

tř-i

tr-oj-e

Num + A

Num + A

Num + B + C

abstract

unit

group

Finnish 'two'

kak-si

kak-si

kah-de-t

Num + A

Num + A

Num + B + C

abstract

unit

group

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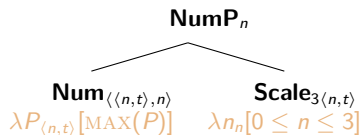
Appendix

Representation of abstract number concepts

Representation of abstract number concepts

- ▶ Number concept expressions realize **two features**: [Num] and [Scale]
- ▶ (22): structure of abstract number concept expressions
- ▶ As in *My favorite number is **three***.

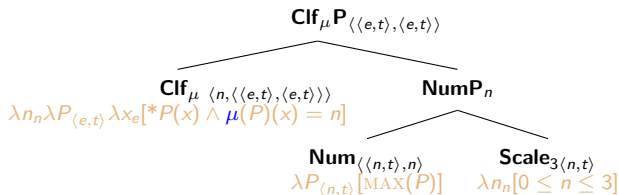
(22)



Representation of numerals as counting devices

- ▶ Classifiers turn abstract number concepts into **counting devices**
- ▶ Two distinct Clf heads:
 - ▶ $\llbracket \mathbf{Clf}_{\text{unit}} \rrbracket = \lambda n_n \lambda P_{\langle e, t \rangle} \lambda x_e [*P(x) \wedge \#(P)(x) = n]$
 - ▶ $\llbracket \mathbf{Clf}_{\text{group}} \rrbracket = \lambda n_n \lambda P_{\langle e, t \rangle} \lambda x_e [*P(x) \wedge |||(P)(x) = n]$
- ▶ (23): representation of unit/group counting numeral structures

(23)



Definitions of $\#$ and $|||$

$$\begin{array}{ll}
 \forall n_n \forall P_{\langle e, t \rangle} \forall x_e [\#(P)(x) = n] & \leftrightarrow \exists Y_{\langle e, t \rangle} [Y \in \text{PAR}_c(x) \ \& \ \forall y_e [y \in Y \rightarrow y \in P \ \& \ |Y| = n]] \\
 \forall n_n \forall P_{\langle e, t \rangle} \forall x_e [|||(P)(x) = n] & \leftrightarrow \exists Y_{\langle e, t \rangle} [Y \in \text{PAR}_c(x) \ \& \ \forall y_e [y \in Y \rightarrow y \in *P \ \& \ |Y| = n]]
 \end{array}$$

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Mandarin and Uzbek

Yucatec Maya and Bwamu

Conclusion

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Unit vs. group counting

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Account

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Conclusion

Appendix

Pattern 1: Mandarin and Uzbek

Mandarin 'three'

san

san **tóu**_{unit}

san **qún**_{group}

Num

Num + **A**

Num + **B**

abstract

unit

group

Uzbek 'two'

ikki

ikki **dona**_{unit}

ikki **guruh**_{group}

Num

Num + **A**

Num + **B**

abstract

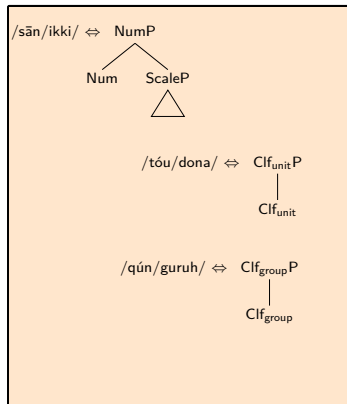
unit

group

Mandarin/Uzbek

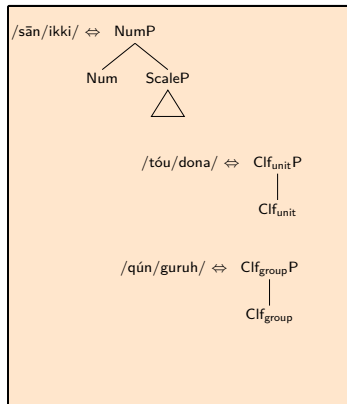
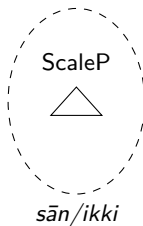
Abstract	Num
Object	Num-A
Group	Num-B

ScaleP



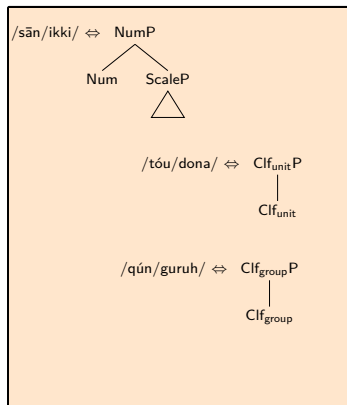
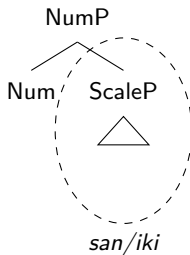
Mandarin/Uzbek

Abstract **Num**
Object Num-A
Group Num-B



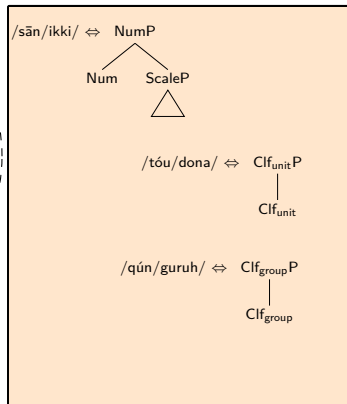
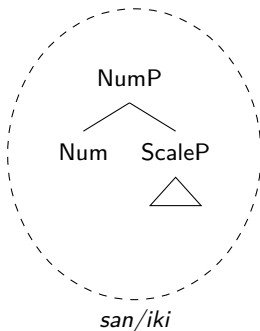
Mandarin/Uzbek

Abstract **Num**
Object Num-A
Group Num-B



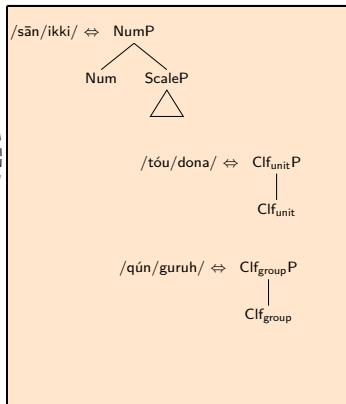
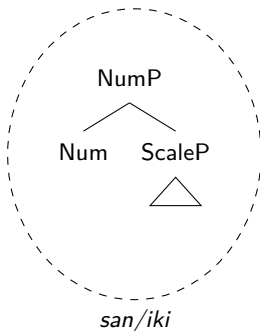
Mandarin/Uzbek

Abstract **Num**
Object Num-A
Group Num-B



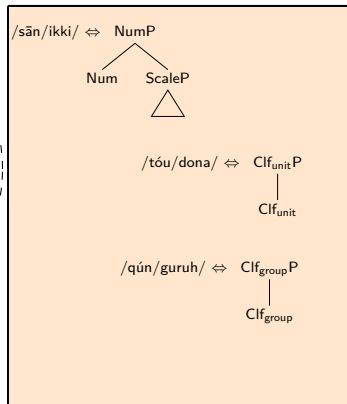
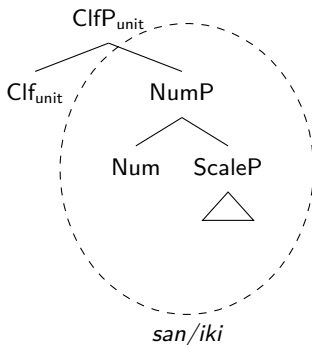
Mandarin/Uzbek

Abstract **Num**
Object Num-A
Group Num-B



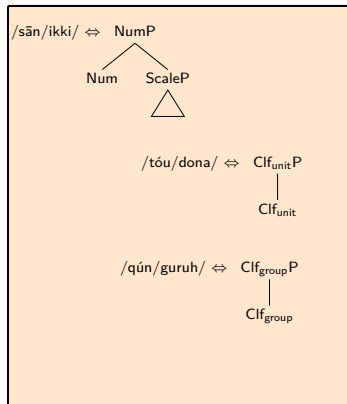
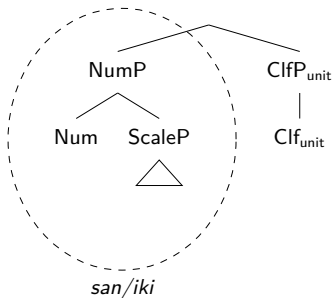
Mandarin/Uzbek

Abstract Num
Object **Num-A**
Group Num-B



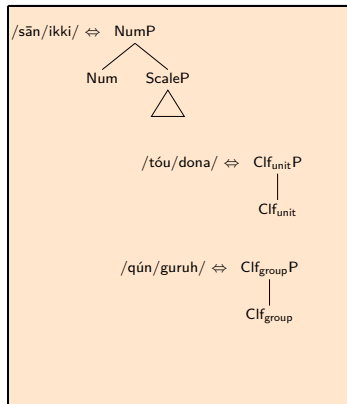
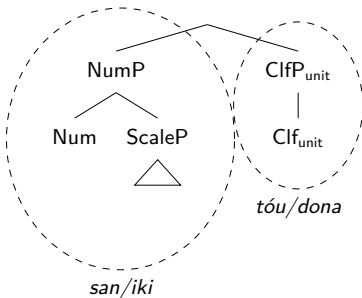
Mandarin/Uzbek

Abstract Num
Object **Num-A**
Group Num-B



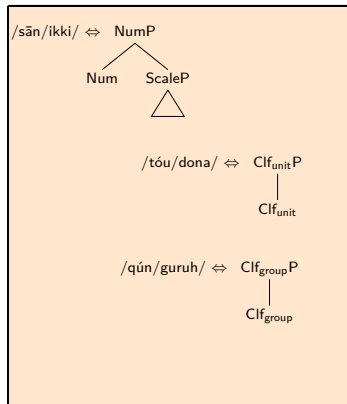
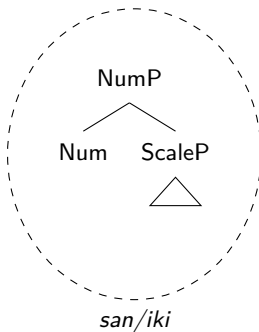
Mandarin/Uzbek

Abstract Num
Object **Num-A**
Group Num-B



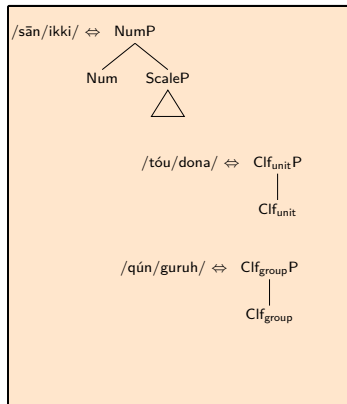
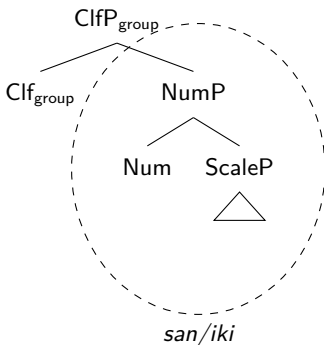
Mandarin/Uzbek

Abstract Num
Object Num-A
Group **Num-B**



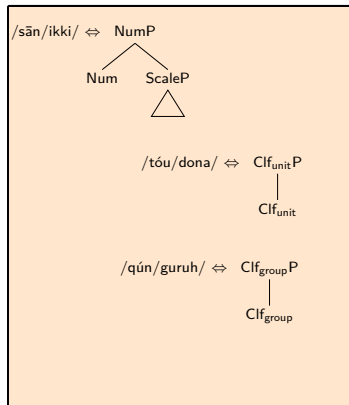
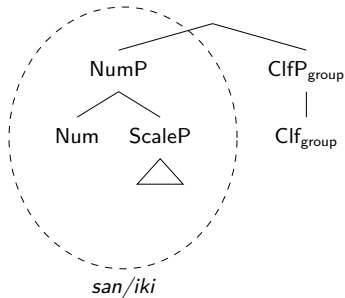
Mandarin/Uzbek

Abstract Num
Object Num-A
Group **Num-B**



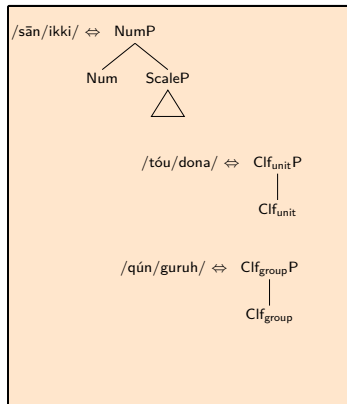
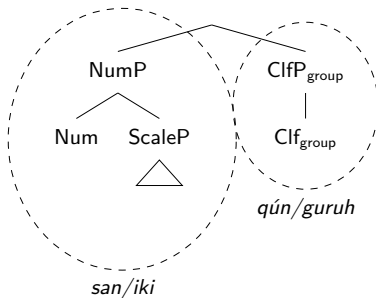
Mandarin/Uzbek

Abstract Num
Object Num-A
Group **Num-B**



Mandarin/Uzbek

Abstract Num
Object Num-A
Group **Num-B**



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Pattern 2: Yucatec Maya and Bwamu

Bwamu 'three'

bo tĩ

bo tĩ

gnũ-nɛ bo tĩ

Num + A

Num + A

Num + A + B

abstract

unit

group

Yucatec Maya 'one'

jum-p'éel

jum-p'éel

jum-p'éel cháach

Num + A

Num + A

Num + A + C

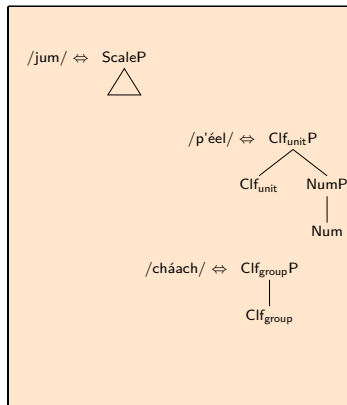
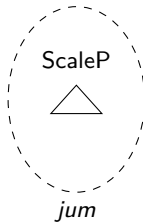
abstract

unit

group

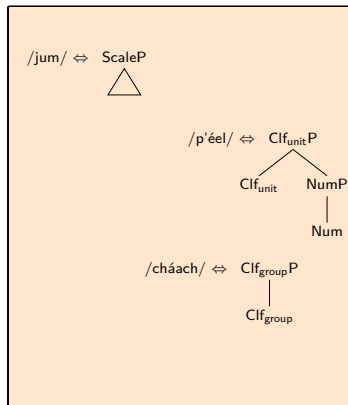
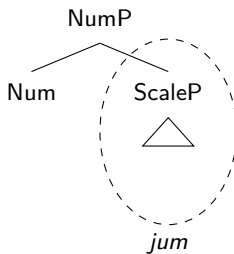
Yucatec

Abstract **Num-A**
Object Num-A
Group Num-A-C



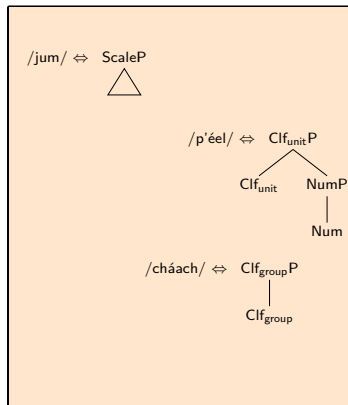
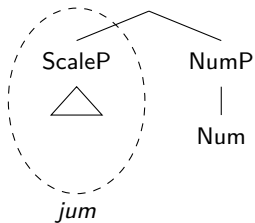
Yucatec

Abstract **Num-A**
 Object Num-A
 Group Num-A-C



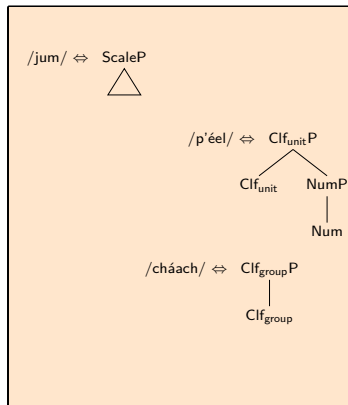
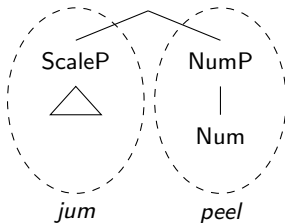
Yucatec

Abstract **Num-A**
 Object Num-A
 Group Num-A-C



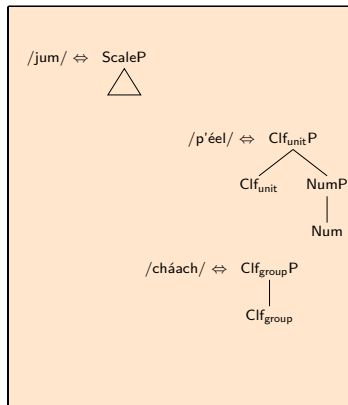
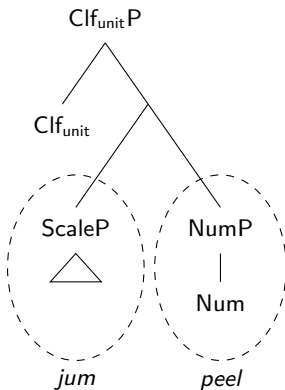
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 Group Num-A-C



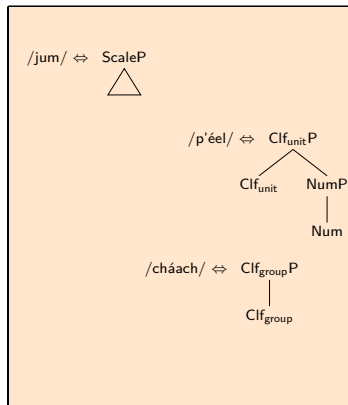
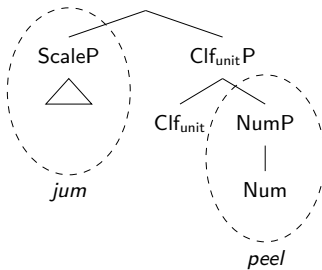
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Abstract Num-A
 Object **Num-A**
 Group Num-A-C



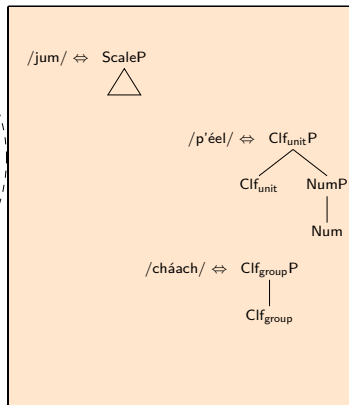
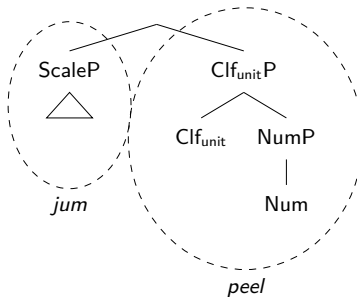
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Abstract Num-A
 Object **Num-A**
 Group Num-A-C



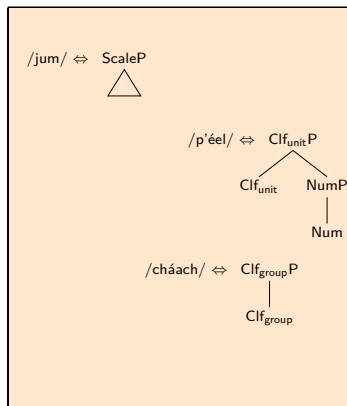
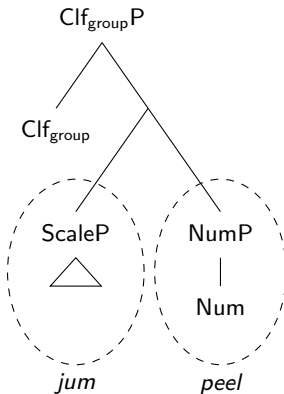
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Abstract Num-A
 Object **Num-A**
 Group Num-A-C



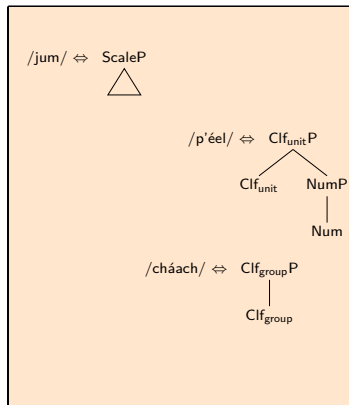
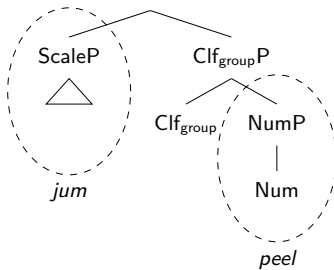
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Abstract Num-A
 Object Num-A
 Group **Num-A-C**



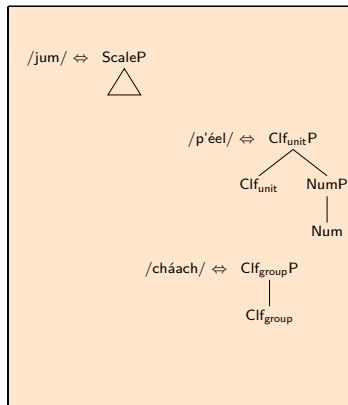
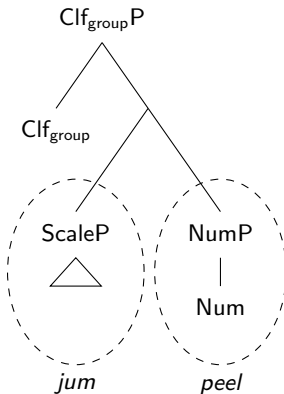
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Abstract Num-A
 Object Num-A
 Group **Num-A-C**



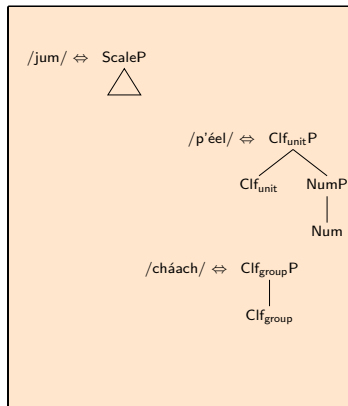
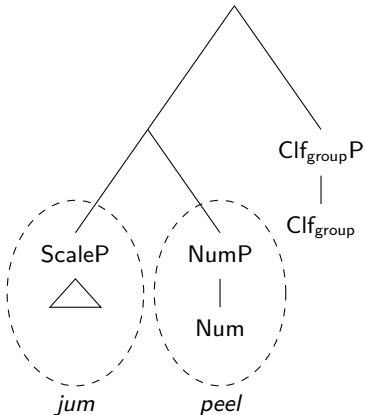
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Abstract Num-A
 Object Num-A
 Group **Num-A-C**



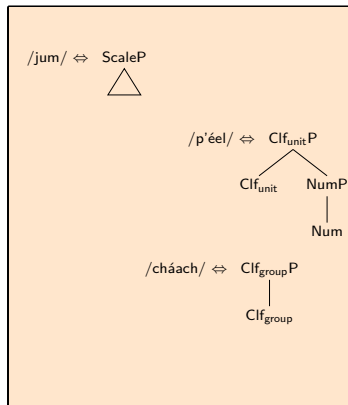
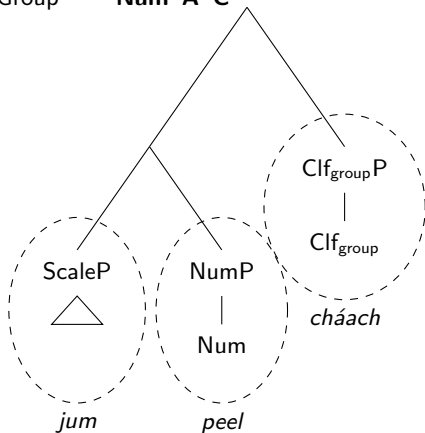
Yucatec

Abstract Num-A
 Object Num-A
 Group **Num-A-C**



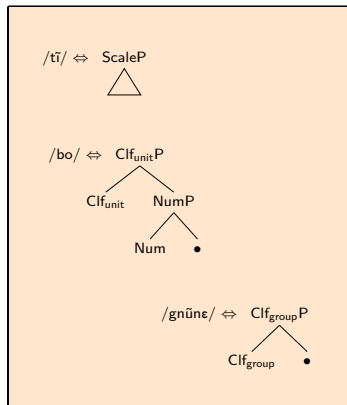
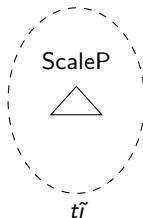
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Abstract Num-A
 Object Num-A
 Group **Num-A-C**



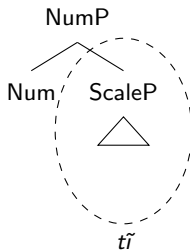
Bwamu

Abstract **Num-A**
 Object Num-A
 Group Num-A-C



Bwamu

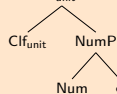
Abstract **Num-A**
 Object Num-A
 Group Num-A-C



$/t\tilde{i}/ \Leftrightarrow$ ScaleP



$/bo/ \Leftrightarrow$ Clf_{unit}P

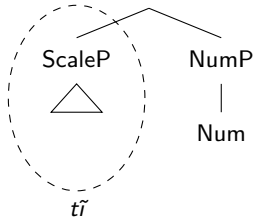


$/gn\tilde{u}\epsilon/ \Leftrightarrow$ Clf_{group}P



Bwamu

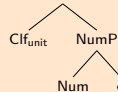
Abstract **Num-A**
 Object Num-A
 Group Num-A-C



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$/bo/ \Leftrightarrow$ Clf_{unit}P

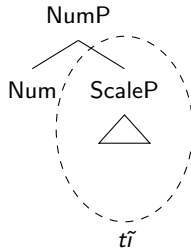


$/gn\tilde{u}\epsilon/ \Leftrightarrow$ Clf_{group}P



Bwamu

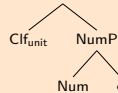
Abstract **Num-A**
 Object Num-A
 Group Num-A-C



$/t\tilde{t}/ \Leftrightarrow \text{ScaleP}$



$/bo/ \Leftrightarrow \text{Clf}_{\text{unit}}\text{P}$

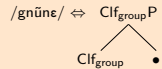
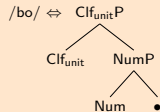
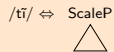
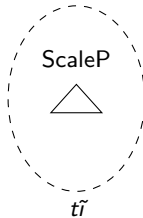
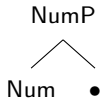


$/gn\tilde{u}\epsilon/ \Leftrightarrow \text{Clf}_{\text{group}}\text{P}$



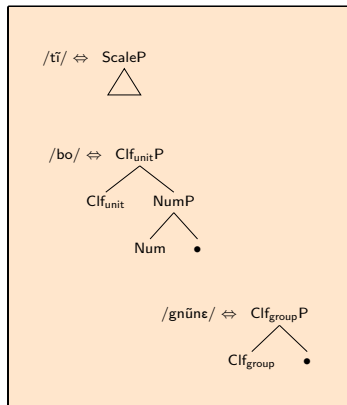
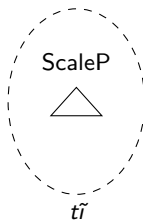
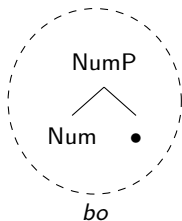
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Abstract **Num-A**
 Object Num-A
 Group Num-A-C



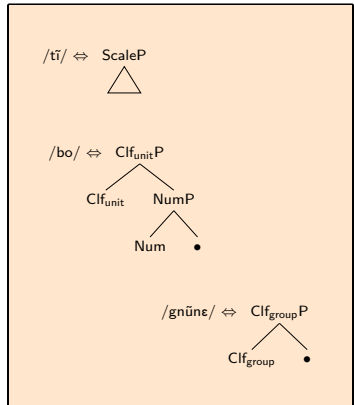
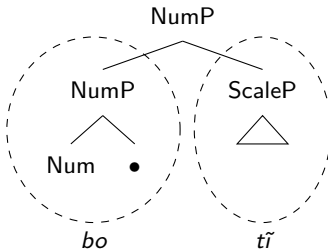
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Abstract **Num-A**
 Object Num-A
 Group Num-A-C



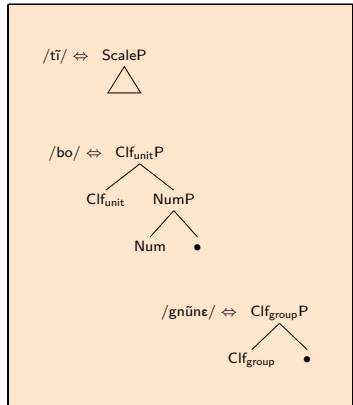
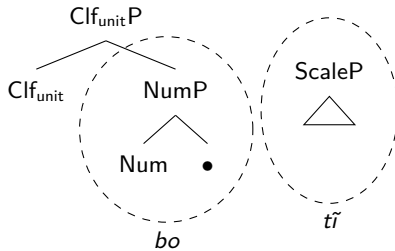
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Abstract **Num-A**
 Object Num-A
 Group Num-A-C



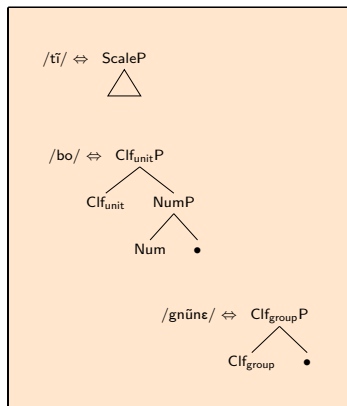
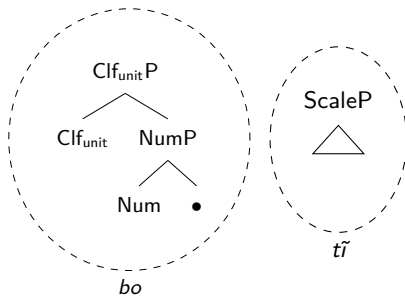
Bwamu

Abstract Num-A
 Object **Num-A**
 Group Num-A-C



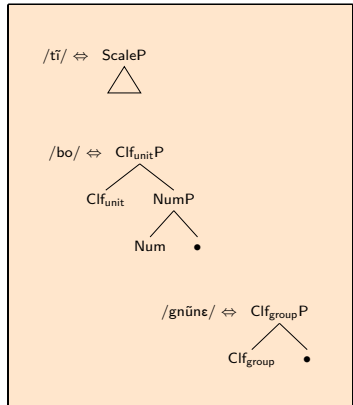
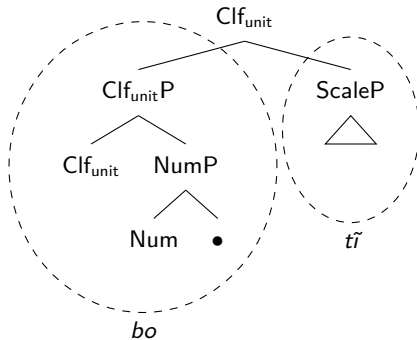
Bwamu

Abstract Num-A
 Object **Num-A**
 Group Num-A-C



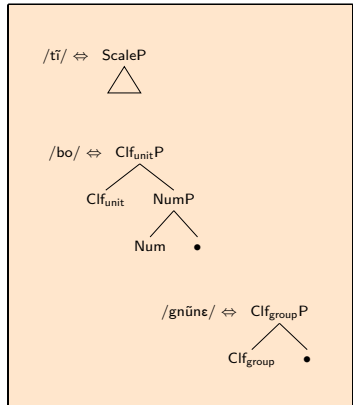
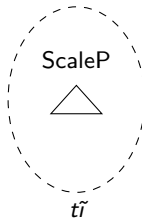
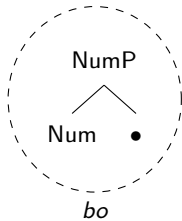
Bwamu

Abstract Num-A
 Object **Num-A**
 Group Num-A-C



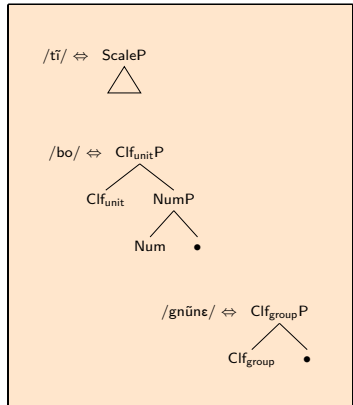
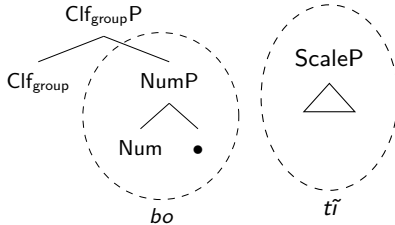
Bwamu

Abstract	Num-A
Object	Num-A
Group	Num-A-C



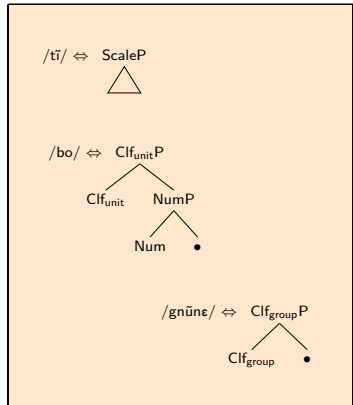
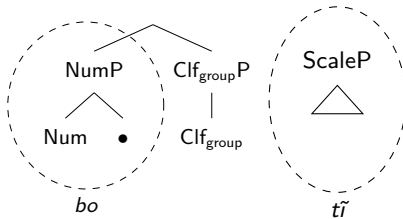
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Abstract	Num-A
Object	Num-A
Group	Num-A-C



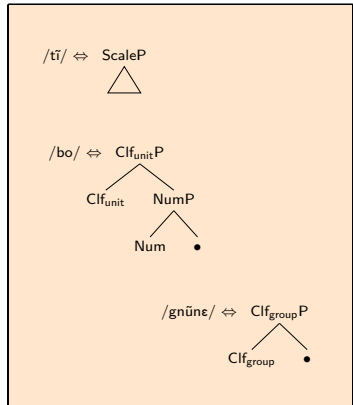
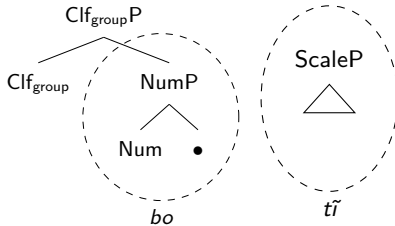
Bwamu

Abstract **Num-A**
 Object Num-A
 Group Num-A-C



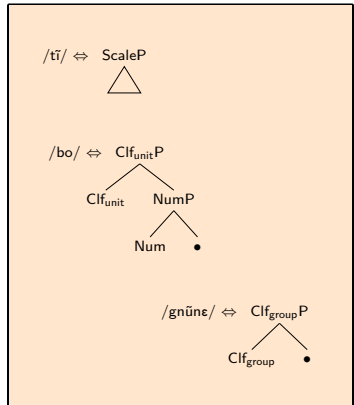
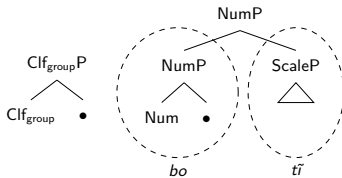
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Abstract **Num-A**
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 Group Num-A-C



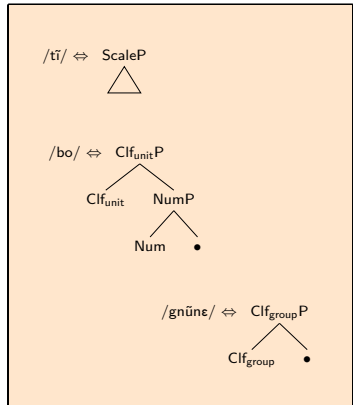
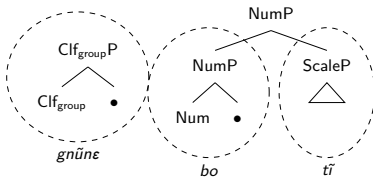
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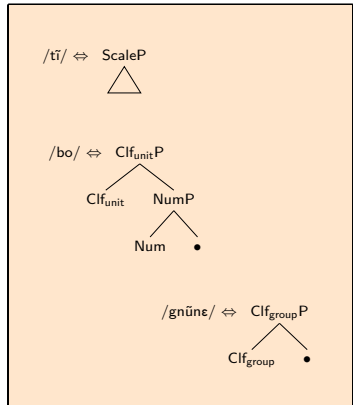
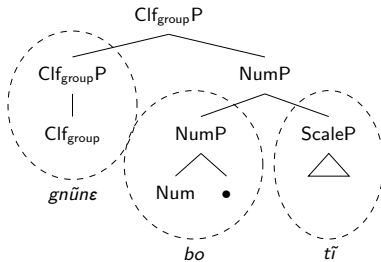
Bwamu

Abstract **Num-A**
 Object Num-A
 Group Num-A-C



Bwamu

Abstract **Num-A**
 Object Num-A
 Group Num-A-C



Outline

Unit vs. group counting

Pattern 1: unit $\rightarrow$ bimorphemic

Pattern 2: unit $\rightarrow$ bimorphemic

group $\rightarrow$ bimorphemic

group $\rightarrow$ tri-morphemic

Abstract use of numerals

Abstract number concepts

Pattern 1: abstract bare *Num*

Pattern 2: abstract *Num+A*

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Conclusion

Taking into account **group counting numeral constructions** provided insights into the following questions:

What is the relation between abstract number concepts and counting expressions?

- ▶ Numeral counting constructions properly contain an **abstract number core**
- ▶ Counting numerals: [CLF [**abstract-number**]]

How does natural language represent abstract number concepts?

- ▶ **Abstract number** concept expressions are **internally complex**.
- ▶ Number concept denoting constituent: [Num Scale]

Thank you!

We would like to thank Éva Dekány, Mouwèrè Bognana, Andreas Hölzl, Olga Kagan and Markus Pöchtrager for data and helpful feedback.

This work was funded by the GAČR project *Classifying Classifiers* (GA24-10073S).

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Appendix

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Appendix: Decomposition of Czech *troje*

Czech *tr-oj-e*

- (24) tř-i
‘three’
- (25) tr-oj-e
‘three groups of’
- (26) tr-oj-í
‘three kinds of’
- (27) tr-oj-it-y
‘triple (hamburger)’
- (28) tr-oj-násobn-ý
‘three time (champion)’

Finnish *kah-de-t*

- (29)
- Alternatively *kahd-e-t* (

Pattern 2: Czech and Finnish

Czech 'three'

tř-i

tř-i

tr-oj-e

Num + A

Num + A

Num + B + C

abstract

unit

group

Finnish 'two'

kak-si

kak-si

kah-de-t

Num + A

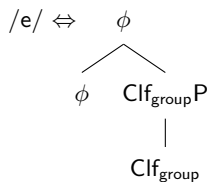
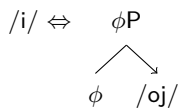
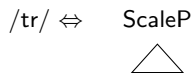
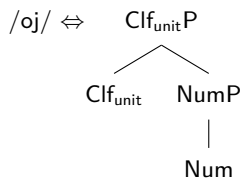
Num + A

Num + B + C

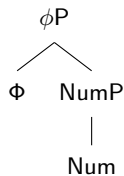
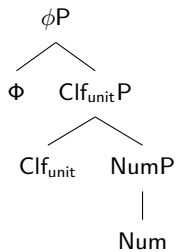
abstract

unit

group

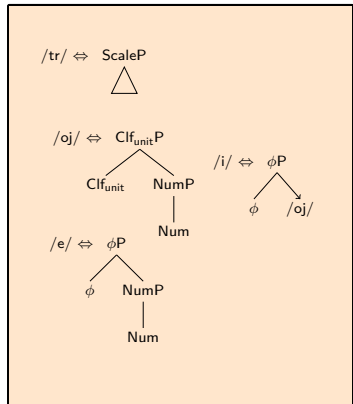
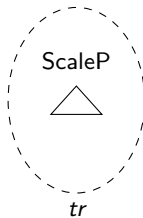


Possible lexicalisations for -i:



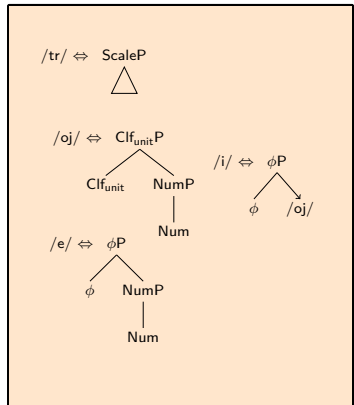
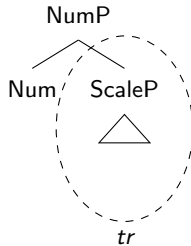
Czech / Finnish

Abstract **Num**
Object Num-A
Group Num-B



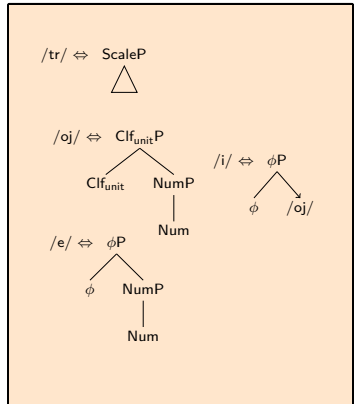
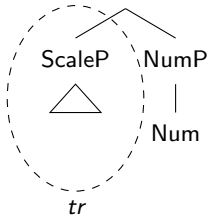
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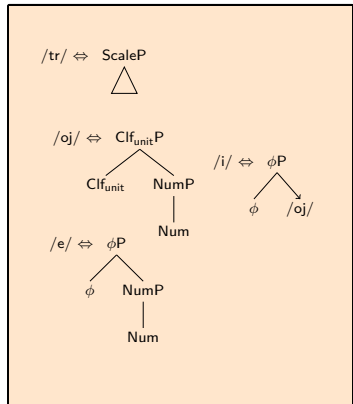
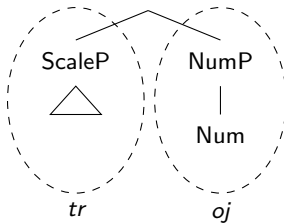
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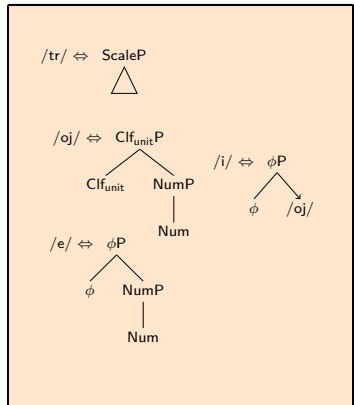
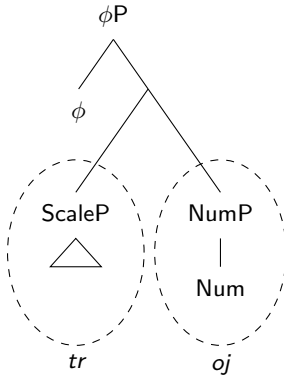
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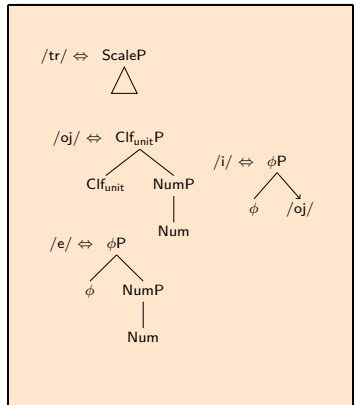
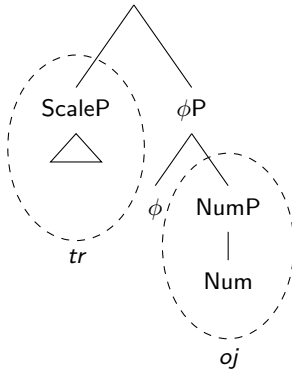
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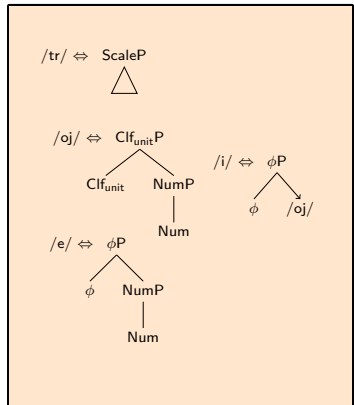
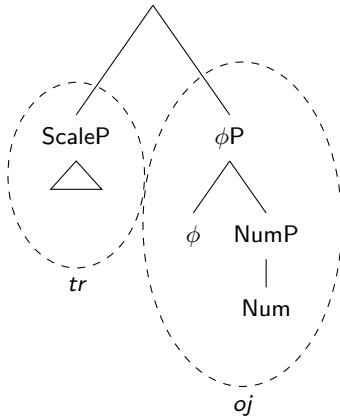
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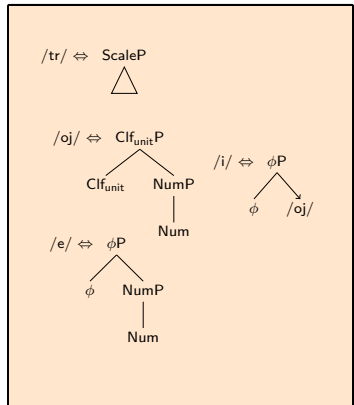
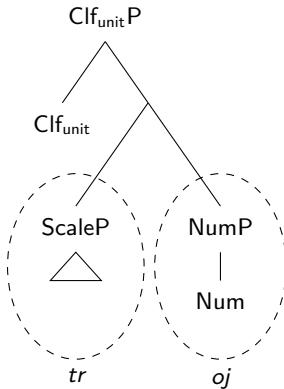
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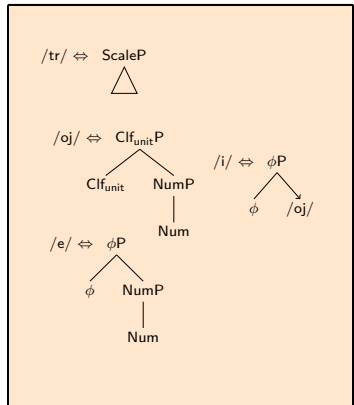
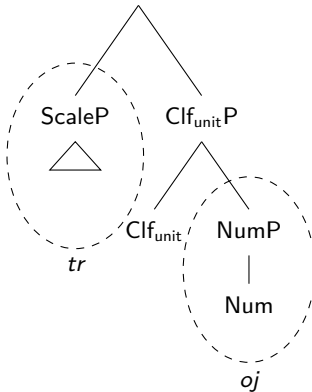
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Abstract Num
 Object **Num-A**
 Group Num-B



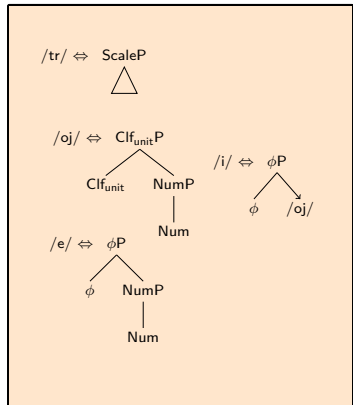
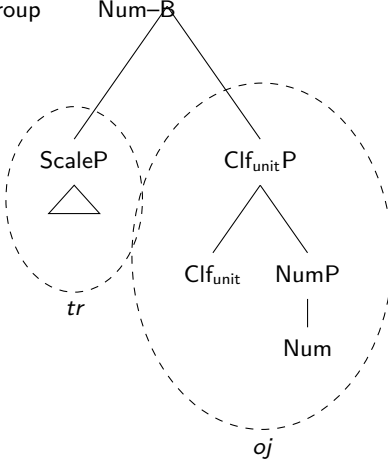
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Group Num-B



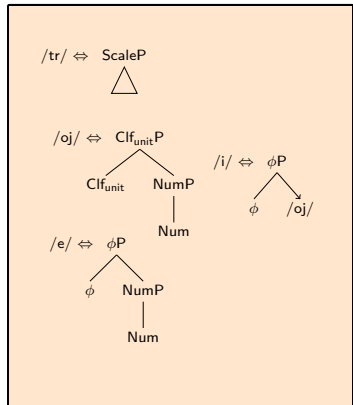
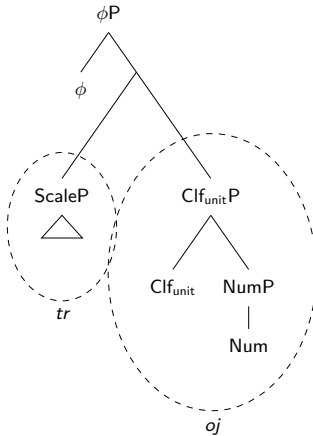
Czech / Finnish

Abstract Num
Object textbfNum-A
Group Num-B



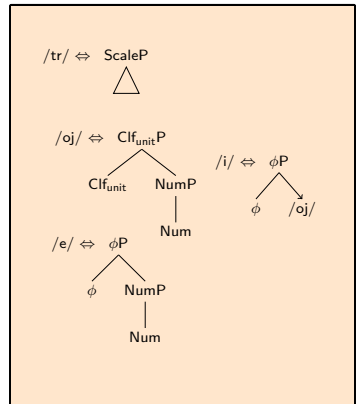
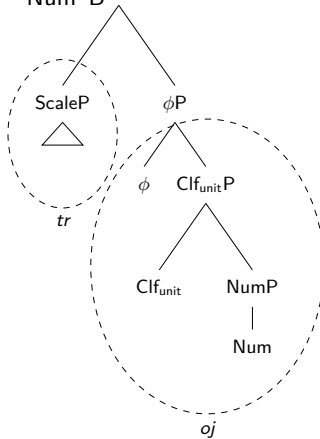
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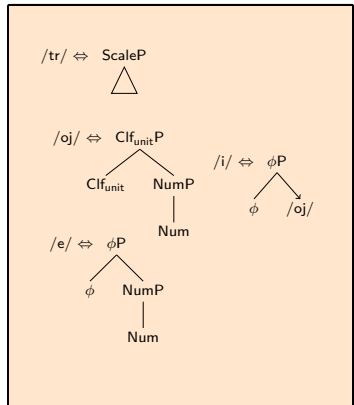
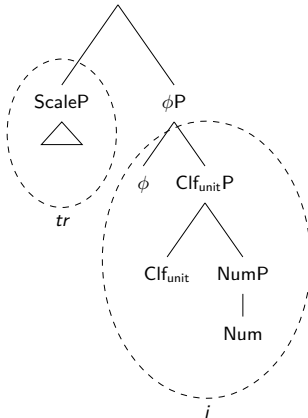
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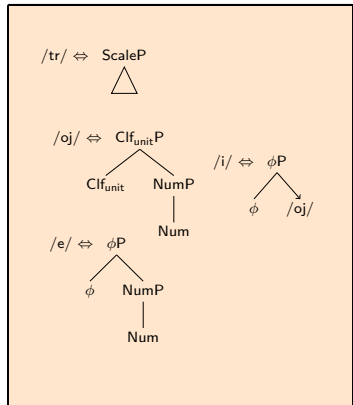
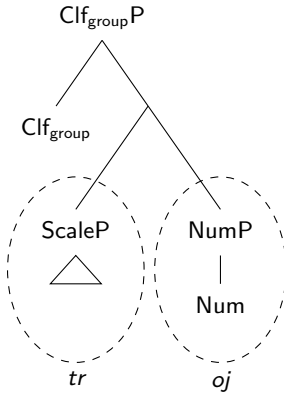
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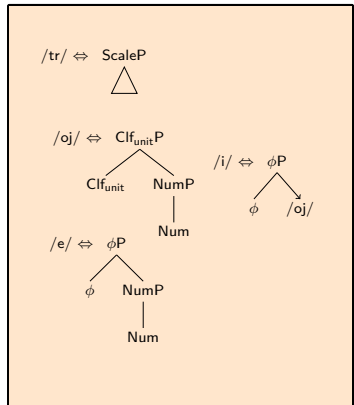
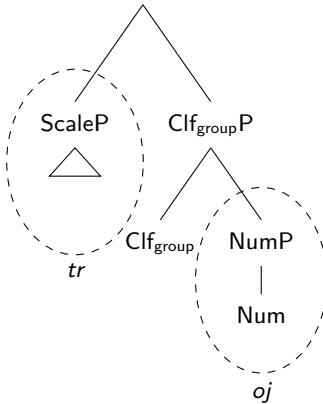
Czech / Finnish

Abstract Num
Object Num-A
Group **Num-C**



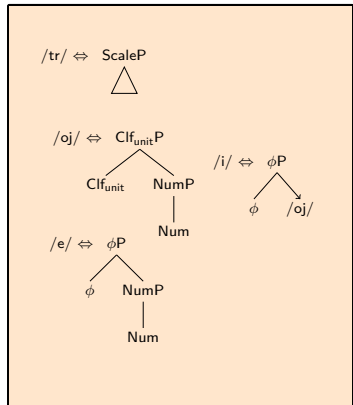
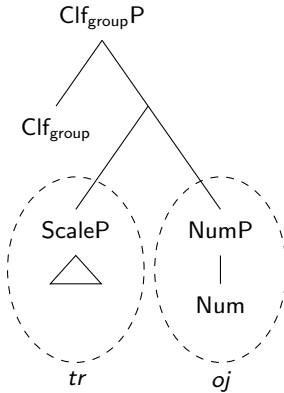
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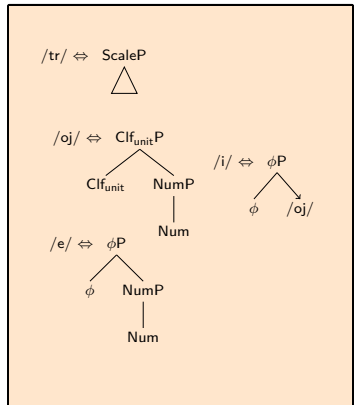
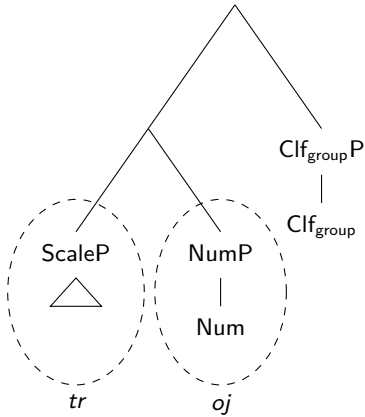
Czech / Finnish

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Czech / Finnish

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Czech / Finnish

Abstract Num
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Group **Num-C**

